Benjamin Holmes

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Education

Massachusetts Institute of Technology **Fall 2023 – Present**

PhD Student

Vassar College **Fall 2019 – Spring 2023**

BA in Computer Science, Departmental Honors

Research Interests

Operating system design for efficient, elastic compute with heterogeneous workloads in the datacenter

Projects/Publications

- *Rethinking Process Snapshots for Near-Warm Serverless Cold Starts* **Benjamin Holmes**, Baltasar Dinis, Lana Honcharuk, Adam Belay, Joshua Fried. (OSDI '26)
- *SEVeriFast: Minimizing the root of trust for fast startup of SEV microVMs.* **Benjamin Holmes**, Jason Waterman, Dan Williams. *Proceedings of the 29th ACM International Conference on Architectural support for Programming Languages and Operating Systems (ASPLOS '24)*
- *KASLR in the age of microVMs.* **Benjamin Holmes**, Jason Waterman, Dan Williams. *Proceedings of the 17th European Conference on Computer Systems (EuroSys '22)*

Awards

Janet Holdeen-Adams Prize for Excellence in Computer Science **Vassar College, 2023**

Sigma Xi Inductee **Vassar College, 2023**

Experience

Teaching Assistant **Fall 2025**

Operating Systems Engineering *Massachusetts Institute of Technology*

Research Assistant **Fall 2023 – Present**

Advisor: Adam Belay *Massachusetts Institute of Technology*

Research Assistant **Summer 2021 – Spring 2023**

Advisors: Jason Waterman and Dan Williams (Virginia Tech) *Vassar College*

Teaching Assistant **Fall 2023**

Operating Systems *Vassar College*

Teaching Assistant **Spring 2021**

Introduction to Computer Systems *Vassar College*

Talks

Guest lecture, CIS 7000, University of Pennsylvania **November 2026**

OSDI, conference talk **July 2026**

ACE Annual Review, Multi-PI/Project Overview/Demonstration (https://acecenter.grainger.illinois.edu/) **October 2025**

ASPLOS, conference talk **April 2024**

Technical Skills

Linux kernel Development	Virtualization	Networked Systems
Performance Engineering	Operating System Design	C, Rust, C++, Go, Python, Java

Relevant Coursework

- Operating Systems
- Computer Networks
- Distributed Systems